

Ehsan Sharafian

— Ph.D. Candidate, Mechanical Engineering | University of Maine

✉ ehsan.sharafian@maine.edu

☎ (207) 631-6979

🌐 ehsansharafian.com

🌐 LinkedIn

🎓 Google Scholar

🐙 GitHub

Dear Hiring Manager,

I am writing to express my strong interest in the [Position title] internship position at [Company name] for Summer 2027. I am a Ph.D. candidate in Mechanical Engineering at the University of Maine. My research focuses on wearable sensing, machine learning, biomechanics, gait rehabilitation, and digital health monitoring. I am particularly interested in [Company name] because of its work in [specific technology/product/research area], which closely aligns with my experience in [2-3 skills most relevant to the role].

During my Ph.D., I have engineered end-to-end wearable solutions, integrating custom mobile applications, IMU sensors, machine learning models, and embedded hardware. I developed a smartphone-deployed gait assessment system using a single IMU to reconstruct foot trajectories, achieving 96% activity classification accuracy and 112 ms processing latency. In another project, I contributed to a haptic feedback system that improved walking speed by up to 30%. These systems have been evaluated with more than 85 participants. Together, these experiences strengthened my ability to design for real users under real-world constraints while adapting system behavior and feedback to individual users. I would bring that same combination of technical depth and hands-on validation experience to [specific team or role at Company name].

Before beginning my Ph.D., I worked as a control system engineer at Sina Robotics, where I developed PLC-based control logic for a telesurgical laparoscopic robotic platform and validated inverse-kinematics algorithms in Simulink. I also built a Python-based user interface and participated in hands-on system testing with laparoscopic surgeons. That experience gave me a sense of the real challenges of developing high-stakes medical and surgical systems.

I believe my background in [relevant technical areas] would allow me to contribute effectively to [specific team/project/product] while continuing to develop my expertise in [relevant area]. I would be excited to bring my experience to [Company name].

Thank you for considering my application. I would welcome the opportunity to discuss how my background and experience could contribute to your team.

Sincerely,

Ehsan Sharafian

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